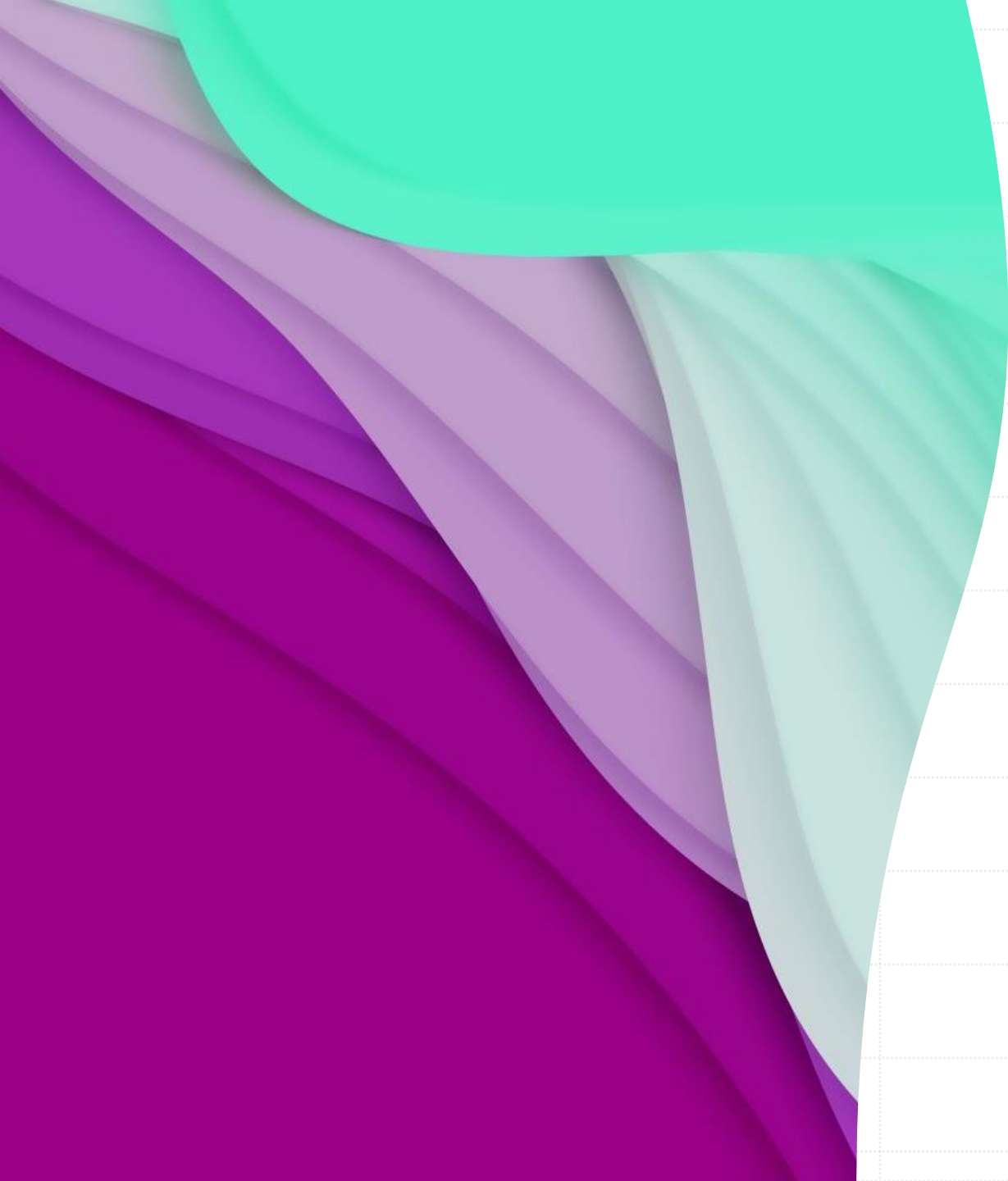






Python Programming Bootcamp – Week 5, Day 01

Introduction to APIs

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Learning Outcomes

- **Understand** the concept of APIs and how they work.
- **Install and use** Python's requests library for HTTP requests.
- **Handle and parse** JSON responses from APIs.
- **Integrate** data from public APIs into Python projects.



What is an API?

- Definition: **Application Programming Interface (API)**
- APIs are mechanisms that enable two software components to communicate with each other using a set of definitions and protocols.
- Examples: Social media integrations, weather data, payment gateways.



Types of APIs

- **Web APIs:** Communicate over HTTP
- **REST vs. SOAP:** REST (**R**epresentational **S**tate **T**ransfer) is simpler and more lightweight.
- **Public vs. Private APIs:** Public APIs are open, while private APIs require authentication.



Real-World API Examples

- **Weather Data:** OpenWeather API
- **Social Media Data:** Twitter API
- **Financial Data:** Exchange rates API



Installing the requests Library

- **Command:** `pip install requests`

Request API> `pip install requests`

- Usage of the requests library for making HTTP requests.



Making a Simple GET Request

- **Example Code:** Fetching data from GitHub's public API

```
import requests
response = requests.get('https://api.github.com')
print(f'Status Code: {response.status_code}')
```




Checking Response Status

- **HTTP Status Codes:**
 - **200 OK**
 - **404 Not Found**
 - **500 Internal Server Error**



Accessing Response Content

```
import requests
response = requests.get('https://api.github.com')
print(f'Status Code: {response.status_code}')
print(f'Contents: {response.content}')
```

Converting Response to JSON

```
import requests
response = requests.get('https://api.github.com')
data = response.json()
print(data)
```

Extracting Specific Data

Example: Extract the `current_user_url` from GitHub's API.

```
import requests
response = requests.get('https://api.github.com')
data = response.json()
current_user_url = data['current_user_url']
print(current_user_url)
```

Handling HTTP Errors

Using **try-except** blocks to handle errors

```
import requests
response = requests.get('https://api.github.com')
try:
    response.raise_for_status()
except requests.exceptions.HTTPError as err:
    print(f"HTTP error occurred: {err}")
```




Query Parameters in API Requests

- Adding **parameters** to refine API requests.
- **Example:** Searching for users on GitHub.

Using Headers in API Requests

```
import requests
headers = {'User-Agent': 'MyApp'}
response = requests.get('https://api.github.com', headers=headers)
data = response.json()
print(data)
```



Fetching Weather Data from an API

- **Example:** Fetching weather data for a city using the OpenWeather API.

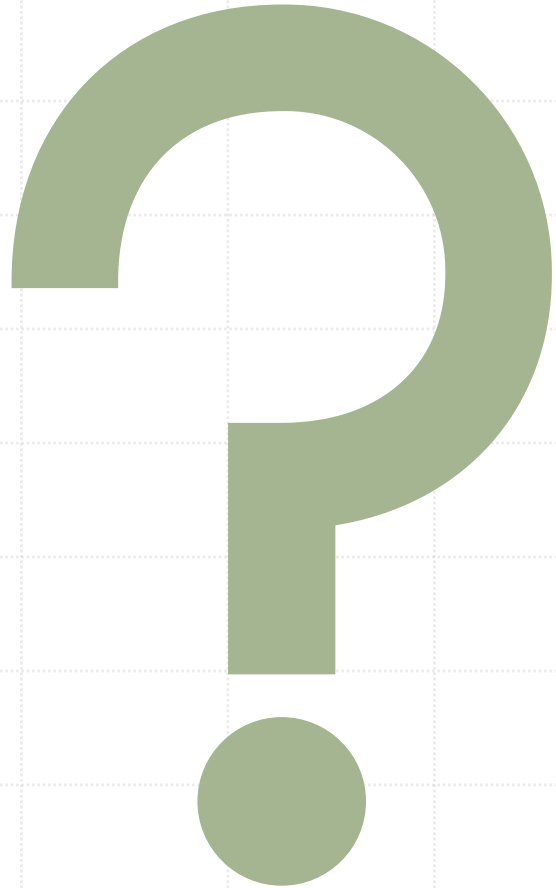
```
import requests
import openweather
city = 'islamabad'
API_key = 'f6e6060d03d7288cd08db756c5ce6726'
response = requests.get(f'http://api.openweathermap.org/geo/1.0/direct?q={city}&appid={API_key}')
data = response.json()
for dic in data:
    for key, value in dic.items():
        print(f'{key} : {value}')
```



```
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lat : 51.4969
lon : -0.0087
country : GB
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lon : 73.0651511
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state : Islamabad Capital Territory
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```



SAMPLE PROJECT



Questions ?